7.2 Analysis of Windowing Effects on Spectral Estimation Using Rectangular, Hamming, and Hanning Windows in MATLAB

```
clc; clear; close all;

% Signal
fs = 1000; N = 256; t = (0:N-1)/fs;
x = sin(2*pi*100*t) + 0.8*sin(2*pi*110*t);
NFFT = 2048; f = (0:NFFT/2-1)*(fs/NFFT);

% Windows
w = {rectwin(N), hamming(N), hann(N)};
names = {'Rectangular','Hamming','Hanning'};

% Processing
for i = 1:3
xw = x(:).*w{i};
X = fft(xw,NFFT);
P{i} = abs(X(1:NFFT/2));
P{i} = P{i}/max(P{i});

end

%% Individual Spectrums
figure;
for i = 1:3
subplot(3,1,i);
plot(f,20*log10(P{i}),'LineWidth',1.5);
title(['Spectrum using ' names{i} ' Window']);
xlim([0 250]); grid on;
end

%% Comparison
```

```matlab
figure;
plot(f,20*log10(P{1}),'k',f,20*log10(P{2}),'r',f,20*log10(P{3}),'b','LineWidth',1.5);
legend(names); title('Window Comparison'); xlim([50 170]); grid on;

%% Window Shapes
figure;
for i = 1:3
subplot(3,1,i); plot(w{i},'LineWidth',1.5);
title([names{i} ' Window']); grid on;
end

%% Peak Detection

for i = 1:3
[pks{i},loc{i}] = findpeaks(P{i},f,'MinPeakHeight',0.3);
fprintf('\n%s Window Frequencies:\n',names{i});
disp(loc{i});
end

%% Peak Marking (Hamming)
figure;
plot(f,20*log10(P{2}),'LineWidth',1.5); hold on;
plot(loc{2},20*log10(pks{2}),'ro','MarkerFaceColor','r');
title('Detected Frequencies (Hamming)'); xlim([50 170]); grid on;

%% Main Peaks
fprintf('\nMain Spectral Peaks:\n');
for i = 1:3
[~,idx] = max(P{i});
fprintf('%s Window Peak = %.2f Hz\n',names{i},f(idx));
end
```

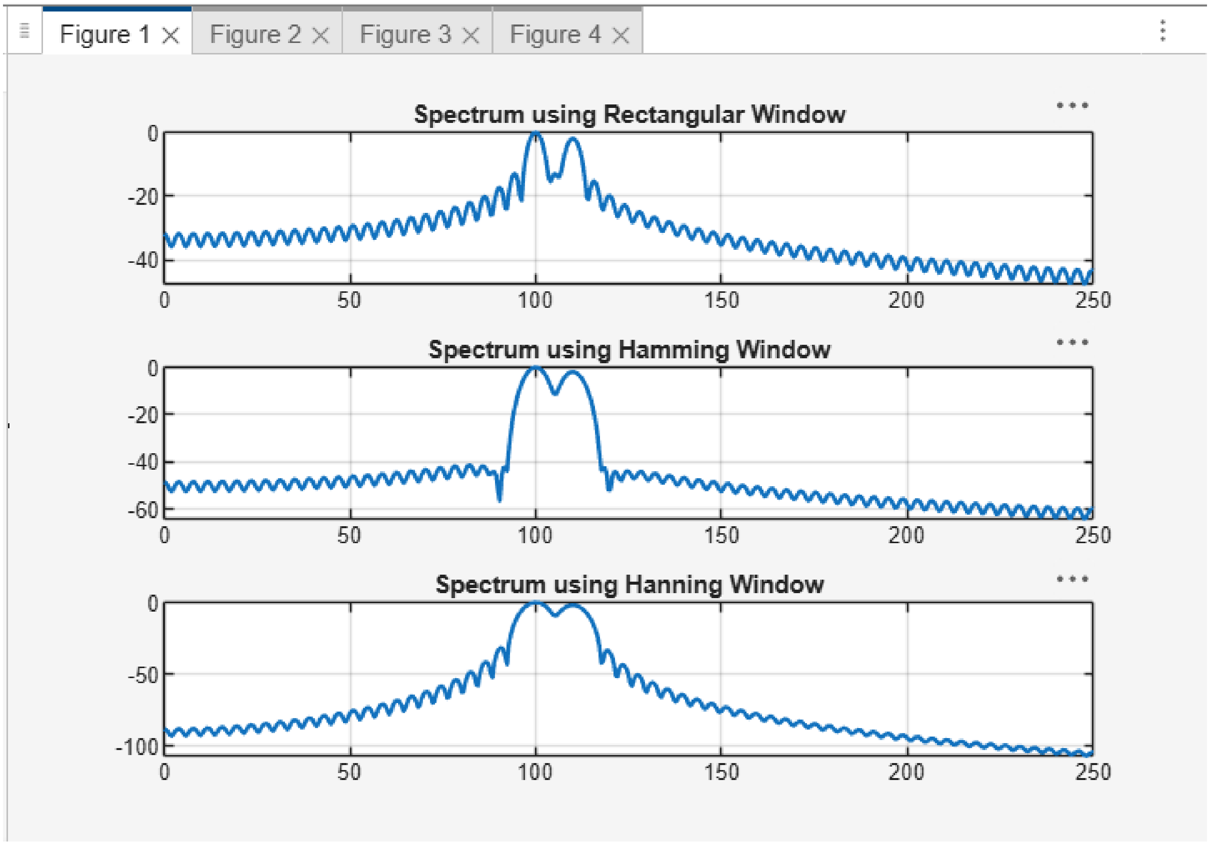

Figure 1
Figure 2
Figure 3
Figure 4
Spectrum using Rectangular Window
Spectrum using Hamming Window
Spectrum using Hanning Window

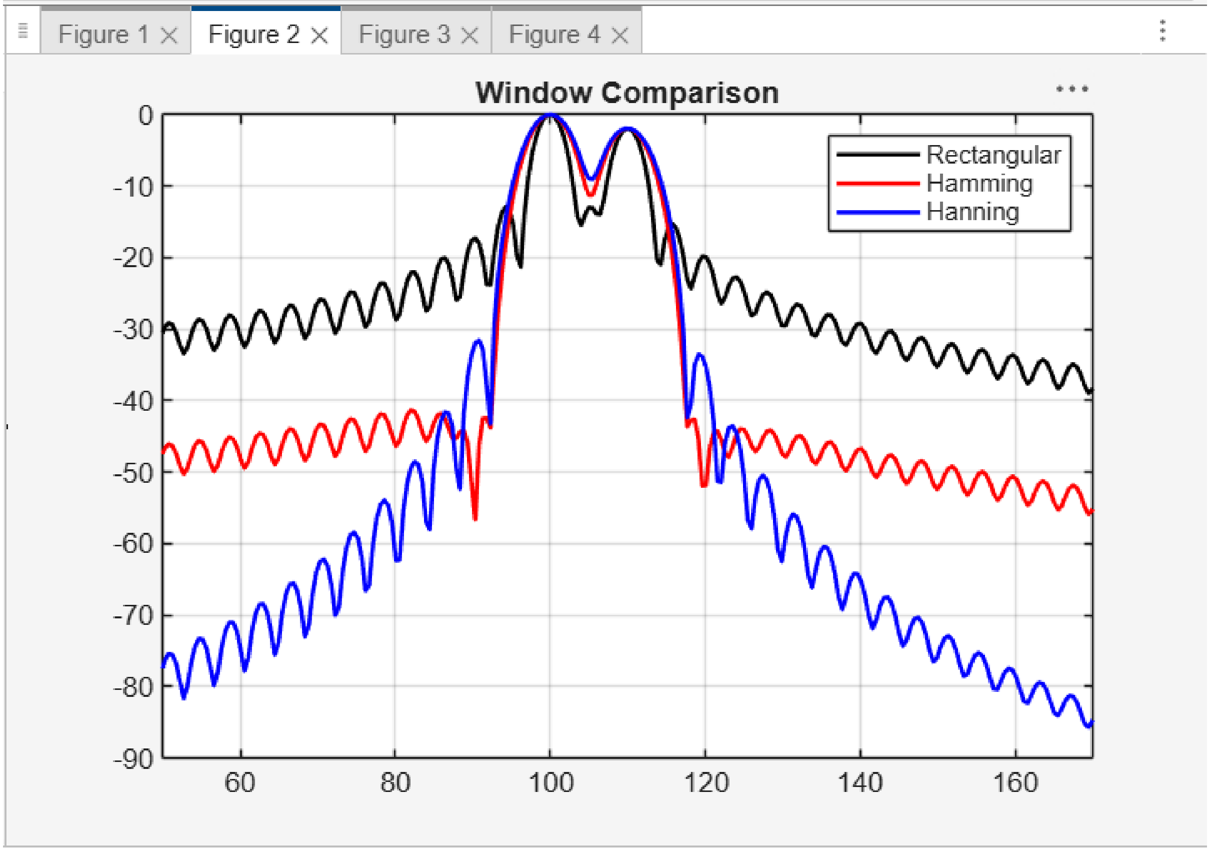

Figure 1
Figure 2
Figure 3
Figure 4
Window Comparison
Rectangular
Hamming
Hanning

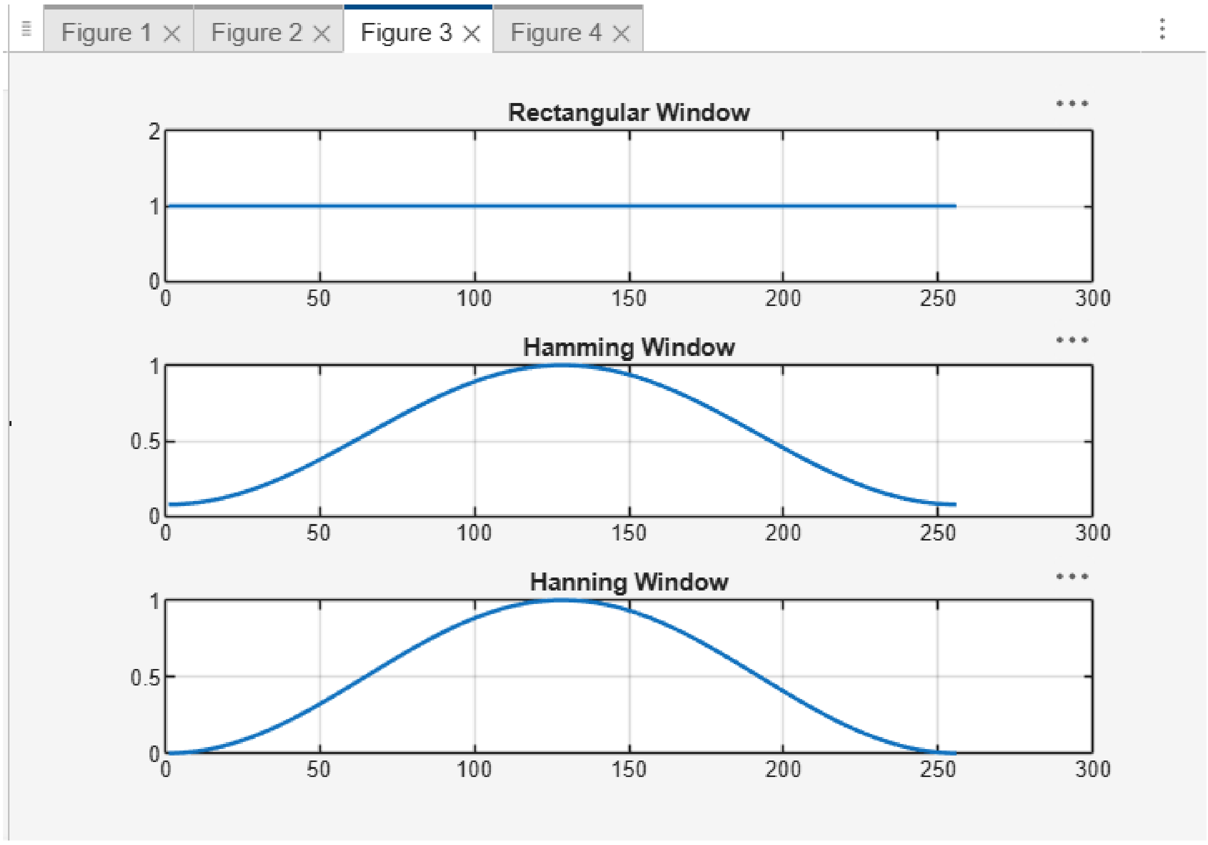
Figure 1
Figure 2
Figure 3
Figure 4
Rectangular Window
Hamming Window
Hanning Window
0 50 100 150 200 250 300
0 0.5 1
0 1 2

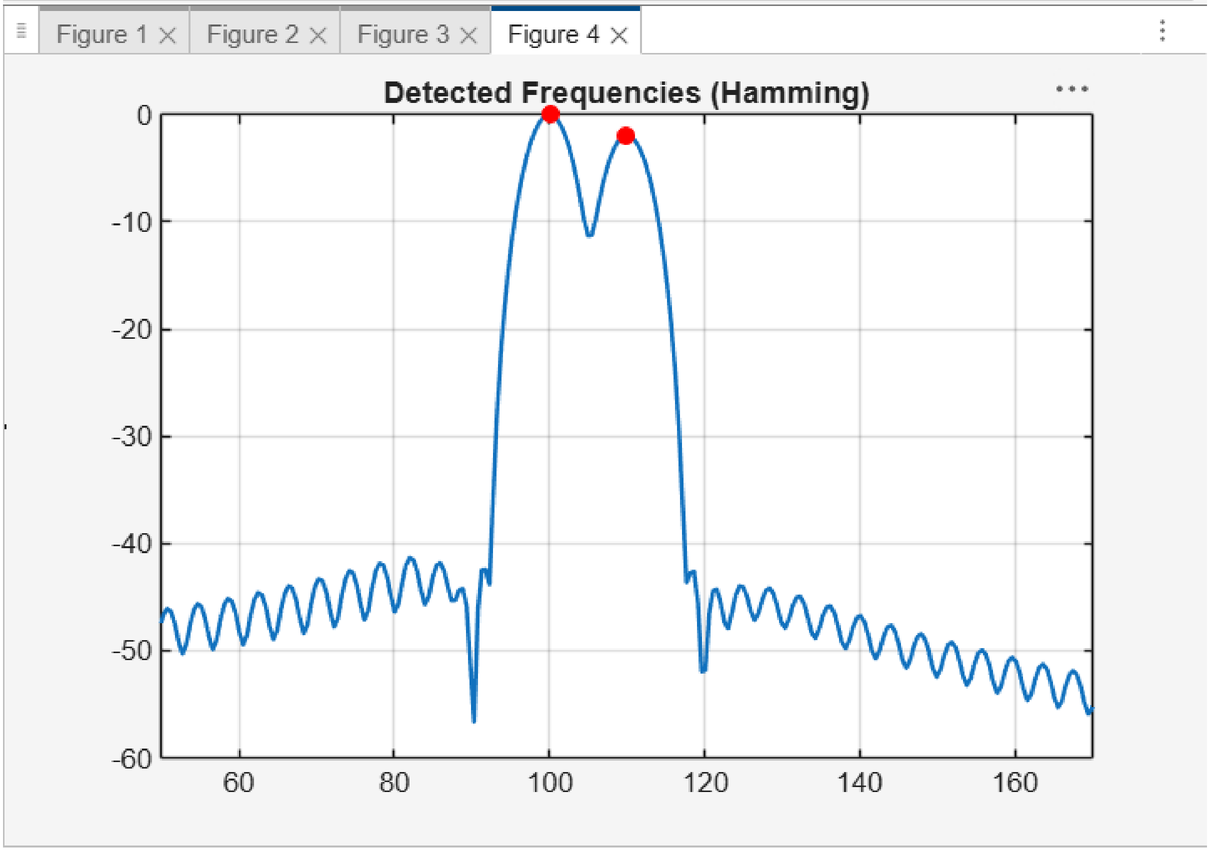
Figure 1
Figure 2
Figure 3
Figure 4
Detected Frequencies (Hamming)
0 -10 -20 -30 -40 -50 -60
60 80 100 120 140 160